

Question: 1

Given a company with 3 departments (HR: 20 users, Sales: 50 users, Tech: 100 users), assign appropriate subnets using VLSM from the 192.168.10.0/24 network and write the subnet address, subnet mask, and usable IP range for each department.

Answer: 1

VLSM Subnet Design Strategy

When using **Variable Length Subnet Masking (VLSM)**, we must always sort the departments from the **largest to the smallest user requirement** to allocate network space efficiently without wasting IP addresses:

- 1. **Tech Department:** 100 users (Needs largest block)
- 2. **Sales Department:** 50 users
- 3. **HR Department:** 20 users (Needs smallest block)

VLSM Department Subnet Allocation Table

Starting from the base network **192.168.10.0/24**, the subnets are systematically carved out as follows:

Department Hierarchy	User Requirement	Subnet Block Size	Assigned CIDR Prefix	Subnet Address	Subnet Mask	Usable IP Range	Broadcast Address
1. Tech	100 Users	128 IPs	/25	192.168.10.0	255.255.255.128	192.168.10.1 to 192.168.10.126	192.168.10.127
2. Sales	50 Users	64 IPs	/26	192.168.10.128	255.255.255.192	192.168.10.129 to 192.168.10.190	192.168.10.191
3. HR	20 Users	32 IPs	/27	192.168.10.192	255.255.255.224	192.168.10.193 to 192.168.10.222	192.168.10.223

Step-by-Step VLSM Calculation Logic

- **Step 1: Tech Department (100 Users)**
 - **Math Check:** Smallest power of 2 greater than 100 is $2^7 = 128$. Subtracting 2 gives 126 usable slots.
 - **CIDR Shift:** Uses 7 host bits ($32 - 7 = 25$), giving a **/25 prefix**.
 - **Block Space:** Takes up IP slots .0 through .127.
- **Step 2: Sales Department (50 Users)**
 - **Math Check:** Smallest power of 2 greater than 50 is $2^6 = 64$. Subtracting 2 gives 62 usable slots.
 - **CIDR Shift:** Uses 6 host bits ($32 - 6 = 26$), giving a **/26 prefix**.
 - **Block Space:** Starts precisely at the next available number .128 and takes up slots .128 through .191.
- **Step 3: HR Department (20 Users)**
 - **Math Check:** Smallest power of 2 greater than 20 is $2^5 = 32$. Subtracting 2 gives 30 usable slots.
 - **CIDR Shift:** Uses 5 host bits ($32 - 5 = 27$), giving a **/27 prefix**.
 - **Block Space:** Starts exactly at the next available number .192 and takes up slots .192 through .223.

Question: 2

You are given three office locations (Ahmedabad, Surat, Rajkot). Each office has three departments: Admin (10 users), Marketing (30 users), and Support (60 users). Design an IP addressing scheme using VLSM for all offices using the 10.0.0.0/22 block, ensuring no overlap and efficient usage. List each subnet with its office, department, subnet mask, and first/last usable IP.

Answer: 2

Multi-Office VLSM Master Table

The base **10.0.0.0/22** block is distributed sequentially from the largest to the smallest department requirements to prevent any IP wastage or overlapping:

Office Location	Department Name	Subnet Address	Subnet Mask	First Usable IP	Last Usable IP
1. Ahmedabad	Support (60 users)	10.0.0.0	255.255.255.192	10.0.0.1	10.0.0.62
	Marketing (30 users)	10.0.0.64	255.255.255.224	10.0.0.65	10.0.0.94
	Admin (10 users)	10.0.0.96	255.255.255.240	10.0.0.97	10.0.0.110
2. Surat	Support (60 users)	10.0.0.128	255.255.255.192	10.0.0.129	10.0.0.190
	Marketing (30 users)	10.0.0.192	255.255.255.224	10.0.0.193	10.0.0.222
	Admin (10 users)	10.0.0.224	255.255.255.240	10.0.0.225	10.0.0.238
3. Rajkot	Support (60 users)	10.0.0.240	255.255.255.192	10.0.0.241	10.0.1.46
	Marketing (30 users)	10.0.1.48	255.255.255.224	10.0.1.49	10.0.1.78
	Admin (10 users)	10.0.1.80	255.255.255.240	10.0.1.81	10.0.1.94

Question: 3

A startup has offices in Mumbai, Pune, and Baroda. Each office will connect to HQ via VPN. Propose a route summarization for all three offices if their assigned subnets are: Mumbai (172.16.10.0/24), Pune (172.16.11.0/24), Baroda (172.16.12.0/24). Write the summarized route and explain why it is efficient.

***Hint:** Find the smallest supernet that includes all three subnets.*

Answer: 3

Summarized Route Calculation

To find the smallest supernet that covers all three offices, we look at the third octet of the IP addresses in binary format:

- **Mumbai:** 172.16.00001010.0 (172.16.10.0/24)
- **Pune:** 172.16.00001011.0 (172.16.11.0/24)
- **Baroda:** 172.16.00001100.0 (172.16.12.0/24)

The first 21 bits of all three addresses match exactly (172.16.00001xxx). The changing bits start from the 22nd bit. Changing the matching bits back to decimal gives a base address of 172.16.8.0 with a prefix length of /22.

- **Final Summarized Route: 172.16.8.0/22**

Why it is Efficient

- **Reduces Routing Table Size:** Instead of forcing the HQ router to maintain three separate routing entries for Mumbai, Pune, and Baroda, it only needs to store one single summarized route.
- **Lowers Router Overhead:** It saves processing power and memory on the HQ router. When data packets travel through the VPN, the router can make quick forwarding decisions based on one supernet rule.
- **Hides Network Instability:** If a network link goes down inside the Baroda office, the summary route at HQ remains active. This stops unnecessary routing update updates from flapping across the entire VPN network.

Question: 4

Given a scenario where the Tech department in Ahmedabad grows from 100 to 200 users, update your previous VLSM plan to accommodate the new size with minimal changes to the rest of the network. List the new subnet details and explain your approach.

Constraint: Do not change the existing subnets for HR or Sales unless absolutely necessary.

Answer: 4

Updated VLSM Subnet Table for Ahmedabad

To support **200 users**, the Tech department's block size must expand from 128 to **256 absolute IP addresses** ($2^8=256$), which changes its prefix from a /25 to a **/24**.

To follow the constraint of leaving the existing **Sales** and **HR** subnets completely untouched, the expanded Tech department is moved to the next available block in the network pool:

Department Name	User Count	Subnet Address	Subnet Mask	First Usable IP	Last Usable IP	Status / Strategy
Sales (Existing)	50 Users	192.168.10.128	255.255.255.192	192.168.10.129	192.168.10.190	Unchanged
HR (Existing)	20 Users	192.168.10.192	255.255.255.224	192.168.10.193	192.168.10.222	Unchanged
Tech (Updated)	200 Users	192.168.11.0	255.255.255.0	192.168.11.1	192.168.11.254	Relocated & Expanded

Explanation of Approach

- **Zero Disruption to Others:** The active subnets for **Sales** (192.168.10.128/26) and **HR** (192.168.10.192/27) are left completely unchanged. This prevents network downtime or re-configuration issues for those existing departments.
- **Accommodating Growth:** Shifting Tech to a /24 network mask provides exactly **254 usable IP addresses**, which easily covers the 200 users and leaves room for future growth.
- **Using the Next Boundary:** Since the 192.168.10.0/24 block contains the active HR and Sales users, the expanded Tech department is cleanly placed at the very next available subnet boundary block: **192.168.11.0**.

Question: 5

Create 5 VLSM subnetting challenge questions inspired by real apps you use (e.g., Zomato delivery zones, Flipkart warehouse locations, or IPL team fan clubs). For each, provide the required number of hosts and a base network address.

Answer: 5

5 Real-App Inspired VLSM Subnetting Challenges

- **Challenge 1: Zomato Delivery Zones (Ahmedabad Hub)**
 - **Base Network Address:** 192.168.50.0/24
 - **Host Requirements:**
 - Satellite Kitchen Zone: **110 hosts**
 - Delivery Rider Hub: **55 hosts**
 - Customer Support Counter: **14 hosts**
- **Challenge 2: Flipkart Fulfillment Center Logistics**
 - **Base Network Address:** 172.16.0.0/22
 - **Host Requirements:**
 - Automated Sorting Floor: **240 hosts**
 - Inventory Tracking Handhelds: **120 hosts**
 - Shipping Office Desktop Nodes: **28 hosts**
- **Challenge 3: IPL Team Fan Club Registry (Mumbai Indians)**
 - **Base Network Address:** 10.5.0.0/23
 - **Host Requirements:**
 - Wankhede Stadium Member Desk: **200 hosts**
 - Online Merchandise Store: **90 hosts**
 - VIP Lounge Local Nodes: **45 hosts**
- **Challenge 4: Swiggy Instamart Warehouse Network**
 - **Base Network Address:** 192.168.100.0/24
 - **Host Requirements:**
 - Cold Storage Sensors: **62 hosts**
 - Billing & Billing Terminals: **30 hosts**
 - Supervisor Workstations: **12 hosts**
- **Challenge 5: JioCinema Live Streaming Regional Gateways**

- **Base Network Address:** 10.20.0.0/21
- **Host Requirements:**
 - Primary Content Delivery Node: **500 hosts**
 - Local Ad Insertion Servers: **210 hosts**
 - Management Console Links: **6 hosts**